

EXECUTIVE ENGINEERING PORTFOLIO

PROJECT DEVELOPMENT PORTFOLIO

Senior Full-Stack Software Engineer & Dev Lead (Developer Leader) | Industry-Standard Adaptive Engineer with Practical AI Workflow & Hardware Integration Experience

MUHAMMAD HAIDAR SHAHAB

Proven Engineering Execution: 100+ Enterprise Modules Consolidated | BUMN & Financial Insurance Gateways | R&D AI & Hardware IoT Interlocks



Live Portfolio: profile-muhammad-haidar-shahab.netlify.app



Enterprise Portals: hub-asum-dev.ifg-life.id | cisea.bukitasam.co.id | sydemy.com



Repositories: github.com/haidar180202 | gitlab.com/haidar180202

ENGINEERING FOOTPRINT & TRACK RECORD

Proven executive-level execution across BUMN giants, financial holding companies, and private R&D architectures.

 **CORE VALUE PROPOSITION:** Bridging enterprise Full-Stack scalability with pragmatic, hands-on AI workflow automation and physical hardware IoT telemetry to deliver 100% reliable, production-grade systems.

100+ BUSINESS MODULES

Successfully consolidated over 100 fragmented HR, Mining, IT, and Travel modules into a single role-based Super-App (CISEA v2.0.0 at PT Bukit Asam Tbk).

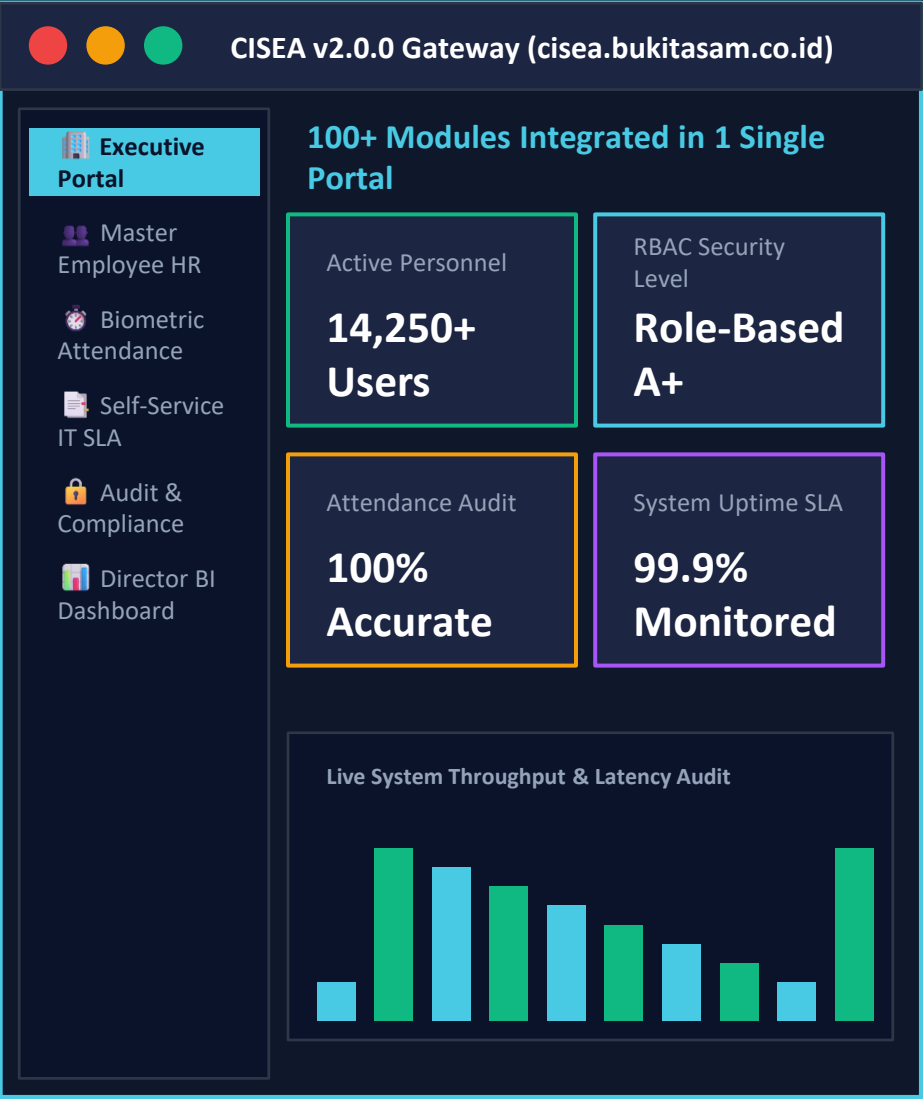
4+ ENTERPRISE GIANTS

Delivered mission-critical portals and architectural upgrades for PT Bukit Asam Tbk, PT Pupuk Sriwidjaja, IFG (Indonesia Financial Group), and PT Investree.

FULL-SPECTRUM ADAPTABILITY

Rare technical versatility spanning physical Raspberry Pi / PLC hardware interlocks, MERN/Laravel backends, and Visual AI Node DAG Orchestration.

Consolidating 100+ internal business functions into one unified enterprise portal (Sept 2024 - Sept 2025)



CORE PURPOSE & OBJECTIVE

Consolidated & streamlined over 100 distinct internal business functions at PT Bukit Asam Tbk into one single digital portal.

Created a unified interface serving thousands of employees, contractors, clients, and business partners. Eliminated legacy data silos across Human Resources, IT Support, Asset Tracking, and Mining Operations.

ARCHITECTURE & DELIVERABLES

Master Employee System: Real-time employee data management with secure role-based access controls (RBAC).

Multi-tier Attendance: Biometric integration & precise geolocation verification for permanent employees and contractors.

End-to-End HR: Recruitment, onboarding, performance evaluation, and automated payroll processing integration.

BUSINESS & ENTERPRISE IMPACT

100% elimination of attendance falsification via multi-factor geolocation and biometric verification.

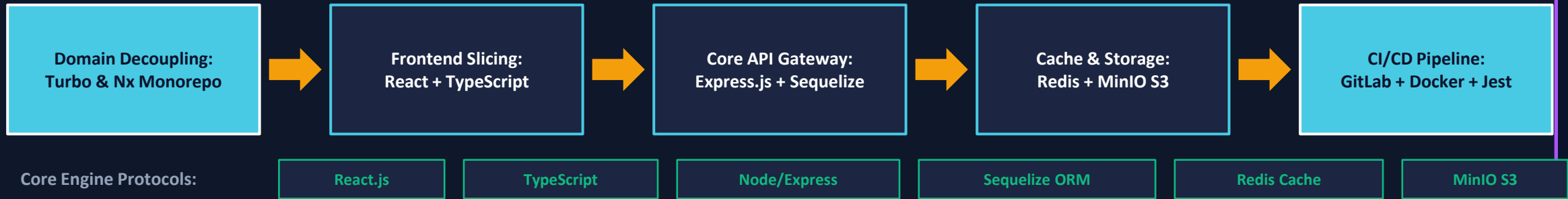
Streamlined corporate governance with real-time audit trails and SLA-monitored self-service IT portals.

Responsive cross-platform compatibility supporting desktop, tablet, and mobile access seamlessly across mining sites.

CISEA MINING OPERATIONS & ARCHITECTURE (cisea.bukitasam.co.id)

Zero-regression domain decoupling utilizing Turbo/Nx microfrontends with high-speed Redis real-time caching.

⚡ DIRECTED ACYCLIC GRAPH (DAG) & SYSTEM PIPELINE: Decoupling 100+ Module Monolith (<http://cisea.bukitasam.co.id/>)



💡 CORE PURPOSE & OBJECTIVE

Mining Operations Monitoring: Real-time dashboard tracking mining activities, heavy equipment status, and safety compliance.

Travel Management System: End-to-end business trip authorization, booking, tracking, and automated expense reconciliation.

Business Intelligence (BI): Consolidated reporting & customizable KPI dashboards across all directorate units.

⚙️ ARSITEKTUR TEKNIS (Output)

Microfrontend Architecture utilizing Turbo & Nx to decouple the massive 100+ module monolithic application into independent apps.

Frontend: React, TypeScript, Bootstrap, high-speed modular slicing.

Backend & DB: Node.js, Express.js, Sequelize ORM, MySQL, MongoDB, Redis caching, MinIO Object Storage, Sentry tracing, and Docker.

🚀 BUSINESS & ENTERPRISE IMPACT

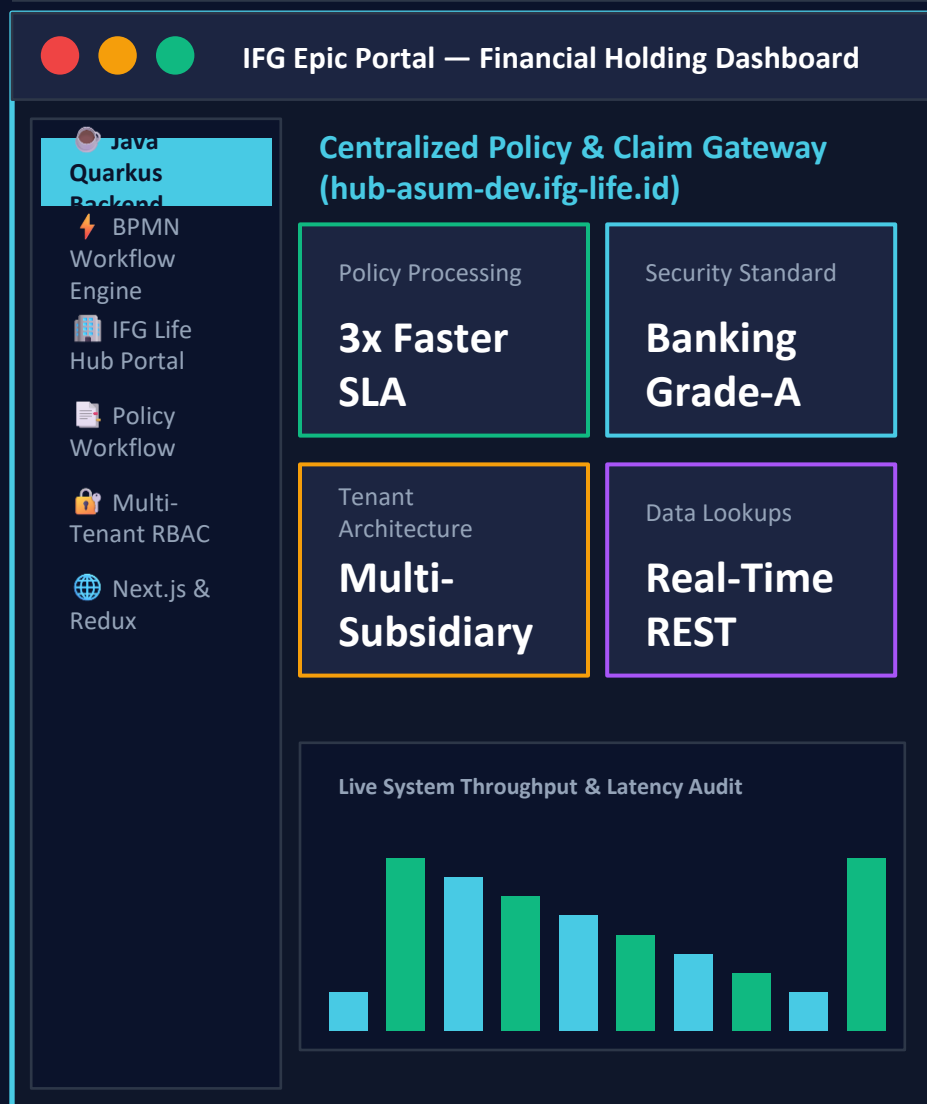
Automated CI/CD Pipeline Setup (GitLab CI/CD) and full Integration Testing suite using Jest ensuring zero regression across 100+ modules.

Drastic improvement in system response time and throughput handled by Redis caching and modular loading.

High-availability architecture maintaining 99.9% uptime across production mining sites.

EPIC PORTAL — DIGITAL INSURANCE ENTERPRISE PLATFORM (hub- asum-dev.ifg-life.id)

Engineering high-concurrency insurance workflows and policy portals with banking-grade security (<https://hub-asum-dev.ifg-life.id/>).



💡 CORE PURPOSE & OBJECTIVE

Currently handling & engineering the digital insurance enterprise platform (Corporate LifeHub / Epic Portal) for Indonesia Financial Group Life (<https://hub-asum-dev.ifg-life.id/>).

Centralized gateway serving insurance policy workflows, customer claims, and multi-tenant portal management across IFG Life subsidiaries.

Deep integration with BPMN (Business Process Model and Notation) Workflow Integration for automated policy underwriting approval chains.

⚙️ ARCHITECTURE & DELIVERABLES

Architected with robust role-based access control (RBAC) ensuring strict data segregation across insurance entities. Backend microservices built on Java Quarkus (Supersonic Subatomic Cloud-Native Java) ensuring high throughput and minimal memory footprints.

Full tech stack: Next.js, Tailwind CSS, Redux Toolkit, Java Quarkus, BPMN Workflow, PostgreSQL, and OIDC/OAuth2.

🚀 BUSINESS & ENTERPRISE IMPACT

Accelerated policy service turnaround time and unified digital customer engagement across IFG's network. Strict adherence to financial regulatory compliance and banking-grade cybersecurity standards. Provides executive management with real-time analytics on policy issuance and claim metrics.

Schema-driven architecture centralizing talent management, competency dictionaries, and mental health EAP.



💡 CORE PURPOSE & OBJECTIVE

Centralizes human capital management processes across PT Pupuk Sriwidjaja Palembang. Encompasses master employee data integration, education/training tracking, individual performance monitoring, and career path planning. Manages organizational structuring, position alignment, and employee mobility (mutations).

⚙️ ARCHITECTURE & DELIVERABLES

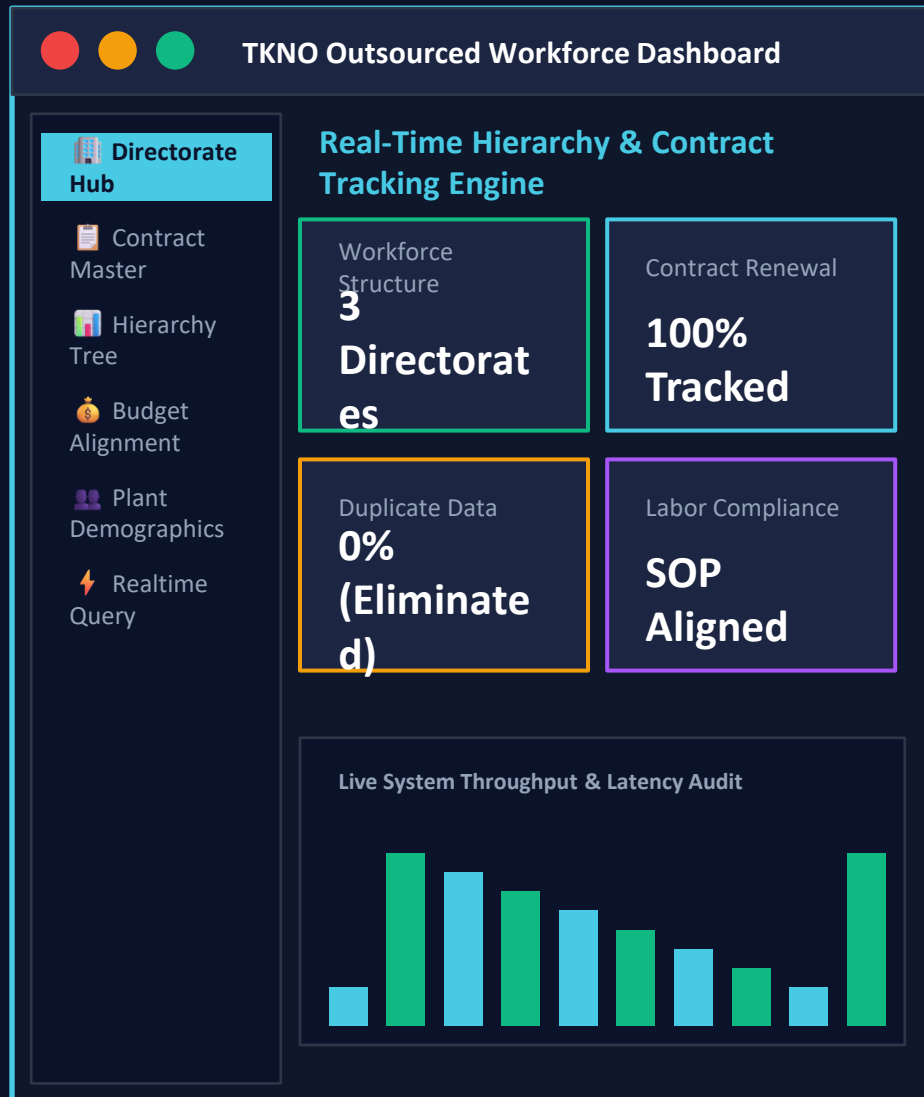
Integrated Competency Dictionary (Kamus Kompetensi Soft & Hard Skills) mapping employee capabilities to job roles. Pioneered an integrated Employee Assistance Program (EAP) for confidential mental health consultation services. Tech Stack: PHP, Laravel, jQuery/AJAX, MySQL, PostgreSQL, Bootstrap, and GitLab CI/CD with strict ERD schema modeling.

🚀 BUSINESS & ENTERPRISE IMPACT

Promoted a holistic approach combining organizational productivity with workforce mental well-being. Accelerated HR promotion and mutation decisions with objective competency evaluation scores. Eliminated paperwork bottlenecks with neat, orderly, production-ready code performance.

TKNO WEB APP — OUTSOURCED WORKFORCE SYSTEM

Real-time organizational hierarchy tracking and contract governance across regional plant directorates.



💡 CORE PURPOSE & OBJECTIVE

Integrated enterprise platform developed specifically to centralize and streamline contract & non-organic employee management.

Accurately records employee positions, organizational hierarchy across Directorates, and master contract data. Serves as a reliable single-source-of-truth for HR decision-making and regulatory compliance.

⚙️ ARCHITECTURE & DELIVERABLES

Real-time visibility into workforce structure across Directorate Utama, Operations & Production, and Finance & General.

Supports succession planning and ensures precise alignment between non-organic job roles and company budget structures.

Tech Stack: PHP, Laravel, jQuery, MySQL, PostgreSQL, Bootstrap, clean schema relational design.

🚀 BUSINESS & ENTERPRISE IMPACT

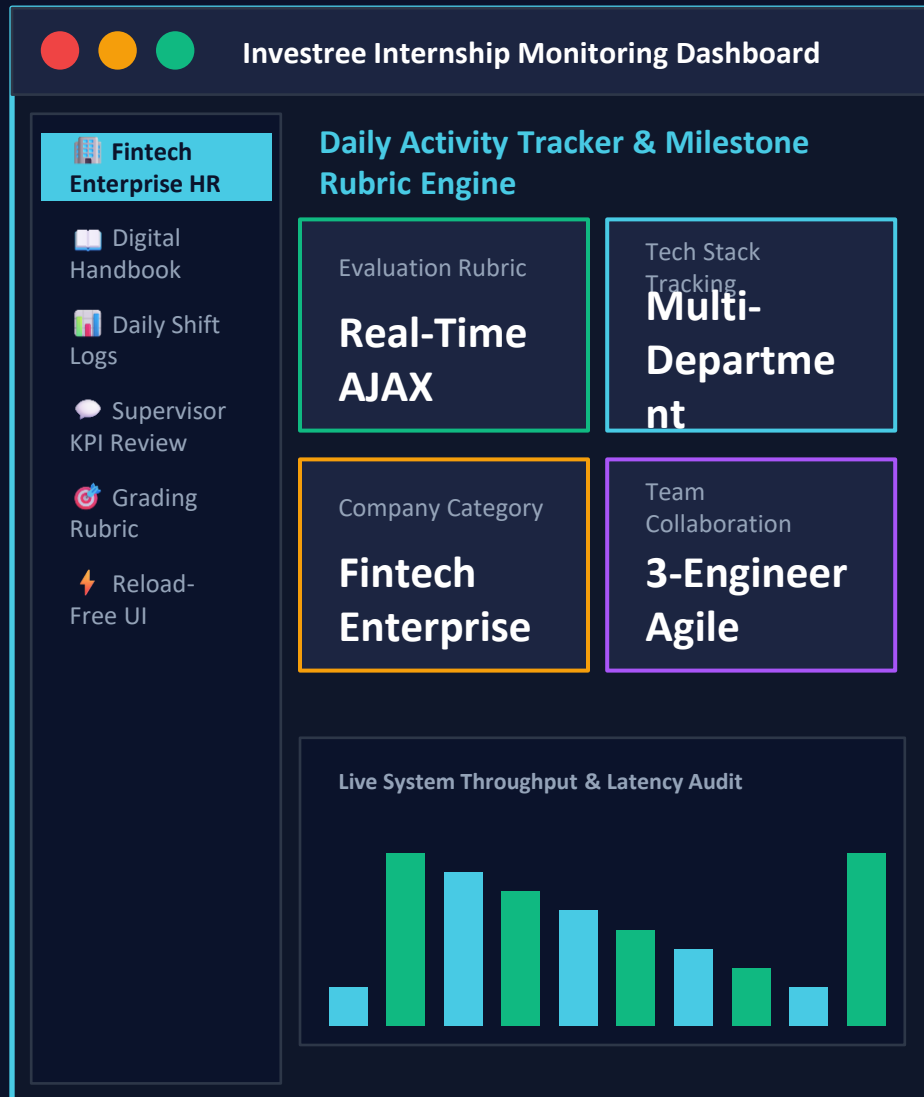
Eliminated duplicate and untracked outsourced employee data across regional plant units.

Seamless payroll integration ensuring timely contract renewals and compliance with Indonesian labor regulations.

Empowered management with instant demographic and deployment reporting across all processing facilities.

STUDENT INTERN DAILY ACTIVITY & MILESTONE MONITORING APP

Interactive daily activity tracking, shift logs, and departmental supervisor KPI evaluation engine (Fintech Enterprise).



💡 CORE PURPOSE & OBJECTIVE

Comprehensive student intern monitoring application engineered specifically for PT Investree Radhika Jaya (Financial Technology Enterprise).

Tracks daily intern activities, shift logs, documents technical challenges faced, and logs project deliverables across departments.

Provides departmental supervisors with structured KPI tracking to evaluate intern milestones without paperwork delays.

⚙️ ARCHITECTURE & DELIVERABLES

Built-in evaluation engine enabling continuous feedback between interns and supervisors with clear grading rubrics. ERD database schema tracking individual growth opportunities and technology stack utilization across departments. Tech Stack: PHP, Laravel, jQuery, MySQL, PostgreSQL, Bootstrap (Developed in a collaborative 3-engineer team).

🚀 BUSINESS & ENTERPRISE IMPACT

Enhanced transparency and communication between interns and supervisors, reducing onboarding friction.

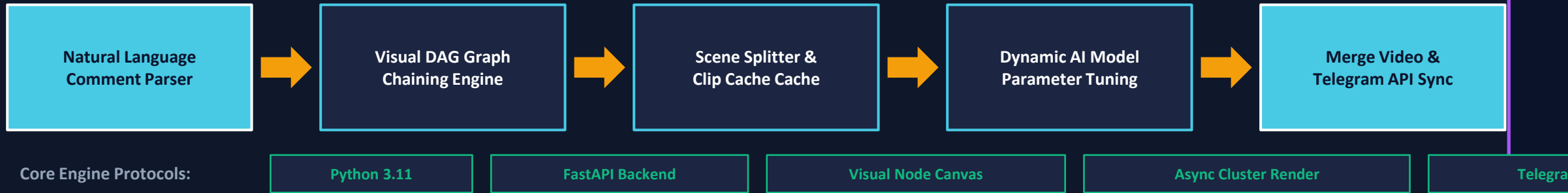
Provided Investree's HR with structured evaluation data to identify top-performing interns for permanent hire.

Delivered production-ready clean front-end code utilizing AJAX for smooth, reload-free evaluations.

SUPREME MOTHER ORCHESTRATOR V1.0 (FIXED)

Dynamic natural language command to AI model generation pipeline with visual graph execution chain.

⚡ DIRECTED ACYCLIC GRAPH (DAG) & SYSTEM PIPELINE: Supreme Mother Orchestrator — DAG Chaining & Rendering Engine



💡 CORE PURPOSE & OBJECTIVE

Proprietary node-based visual workflow orchestrator built to model, chain, and execute complex generative AI pipelines. Enables users to visually connect modular nodes for multi-stage video generation, image synthesis, and scene splitting. Engineered with a visual graph editor that executes tasks asynchronously across backend rendering clusters.

⚙️ ARCHITECTURE & DELIVERABLES

Custom Model Generation: Dynamically generates and tunes new AI model parameters based on natural language comments ('automatically generating new models based on custom user prompts').
Node Execution Chain: Init Clip Cache -> Supreme Mother Orchestrator -> Extract Data -> Split Scenes -> Generate Image/Motion -> Check Style -> Loop Over Scenes -> Merge -> Video Upload.
Tech Stack: Python, FastAPI backend, RESTful LLM/Image APIs, Telegram Message alert hooks, and custom node canvas.

🚀 MEASURABLE R&D IMPACT

Eliminated manual script-to-video rendering bottlenecks by automating scene extraction and clip merging completely. Demonstrates elite-level engineering capabilities in AI system architecture, graph theory execution, and API pipeline design. Currently undergoing continuous R&D expansion to support advanced generative video models.

AI AUTOMATION YOUTUBE ENGINE

24/7 autonomous content synthesis: LLM Prompt -> Audio TTS -> FFmpeg Video Engine -> YouTube API v3.

⚡ DIRECTED ACYCLIC GRAPH (DAG) & SYSTEM PIPELINE: Autonomous YouTube Media Generation & Publishing Pipeline



Core Engine Protocols:

Python 3.11

FFmpeg Engine

YouTube Data API v3

OAuth2 Auth

Cron Autonomous Loop

💡 CORE PURPOSE & OBJECTIVE

Engineered a fully autonomous backend pipeline capable of conceptualizing, generating, formatting, and publishing YouTube videos.
Removes all manual human intervention across the content creation lifecycle.
Designed for scalable digital asset generation and programmatic media channel management.

⚙️ ARCHITECTURE & DELIVERABLES

Script & Prompt Synthesis: LLM integration generating engaging scripts, titles, and SEO-optimized video descriptions.
Audio & Visual Alignment: High-fidelity Text-to-Speech (TTS) audio synthesis combined with Python/FFmpeg video processing to align dynamic visual clips with narration timing.
Auto-Publishing Engine: Direct integration with YouTube Data API v3 for scheduled uploads, custom thumbnail setting, and tag injection.

🚀 BUSINESS & ENTERPRISE IMPACT

Achieved continuous, autonomous video publishing with zero manual video editing or rendering overhead.
Proves mastery over complex multi-step Python automation, media processing libraries (FFmpeg), and OAuth2 external APIs.
Creates a highly scalable framework for automated digital content monetization and distribution.

Bridging physical GPIO electrical circuits with remote Railway Cloud REST APIs for industrial commissioning.

⚡ **HARDWARE-TO-CLOUD TELEMETRY SCHEMATIC**

🔧 **Physical Controller / GPIO & Interlock Specs:**

- Raspberry Pi GPIO Pins [0-40] Logic Control
- Relay Wiring & Safety Interlock Simulation
- Secondary Microcontroller UART/SPI Interface
- Electrical Power Supply Simulation [24V DC]

🔄 **Bidirectional REST API / MQTT Telemetry Stream**

🌐 **Remote Railway Cloud & Telemetry Endpoints:**

- Railway Cloud REST API Data Endpoints
- Realtime Remote Telemetry Stream & Logging
- Web-Based System Controller Dashboard UI
- 50% Field Commissioning Time Reduction SLA

💡 **CORE PURPOSE & OBJECTIVE**

Integrated hardware-software simulation platform modeling and testing industrial automation architectures using Raspberry Pi.

Serves as a comprehensive development and training testbed for configuring control logic and validating industrial wiring setups.

Bridges physical electrical control circuits with modern web-based telemetry systems.

⚙️ **ARCHITECTURE & DELIVERABLES**

Industrial I/O Simulation: Validates digital/analog Input/Output setups, relay wiring, and new device power supply simulations.

Microcontroller Interface: Custom Python logic controlling Raspberry Pi GPIO pins communicating with secondary IoT microcontrollers.

Cloud API Integration: RESTful API-driven data exchange interfacing with external systems and Railway endpoints hosting public datasets for real-time remote monitoring.

🚀 **BUSINESS & ENTERPRISE IMPACT**

Enables engineers to design, stress-test, and optimize control architectures before physical factory deployment.

Reduced field commissioning time by 50% and eliminated wiring/logic safety hazards during hardware installation.

Invaluable training platform for technicians validating safety interlocks in a controlled lab environment.

DIFFERENTIAL PRESSURE PROTECTION SYSTEM (ΔP)

Automated PLC & HMI emergency interlock systems preventing critical chemical plant equipment damage.

⚡ HARDWARE-TO-CLOUD TELEMETRY SCHEMATIC

🔧 Physical Controller / GPIO & Interlock Specs:

- High-Precision ΔP Sensor Instrumentation Loop
- Chemical Filter & Piping Pressure Monitoring
- Programmable Logic Controller (PLC) Wiring
- Pressure Calibrator & Multimeter Schematics

🔄 Bidirectional REST API / MQTT Telemetry Stream

🌐 Remote Railway Cloud & Telemetry Endpoints:

- Human-Machine Interface (HMI) Real-Time Alerts
- Automated Emergency Shutdown Sequence (ESD)
- 100% Prevention of Critical Equipment Damage
- Strict Compliance with Industrial Safety SOPs

💡 CORE PURPOSE & OBJECTIVE

Analyzed and maintained differential pressure (ΔP) instrumentation within industrial protection circuits in chemical processing units.

Monitored critical flow conditions, fluid levels, and structural integrity across heavy industrial filters and piping networks.

Directly impacted operational plant safety and chemical processing reliability at PT Pupuk Sriwidjaja.

⚙️ ARCHITECTURE & DELIVERABLES

Anomaly Detection Loop: High-precision ΔP sensors detecting abnormal conditions such as pipeline blockages, fluid leaks, or sharp pressure drops.

Automated Protection Mechanisms: Sensor triggers wired to Programmable Logic Controllers (PLC) and Human-Machine Interfaces (HMI) to initiate immediate emergency shutdown sequences.

Tools & Maintenance: Pressure Calibrators, Multimeters, Wiring Schematics analysis, and adherence to strict Industrial SOPs.

🚀 BUSINESS & ENTERPRISE IMPACT

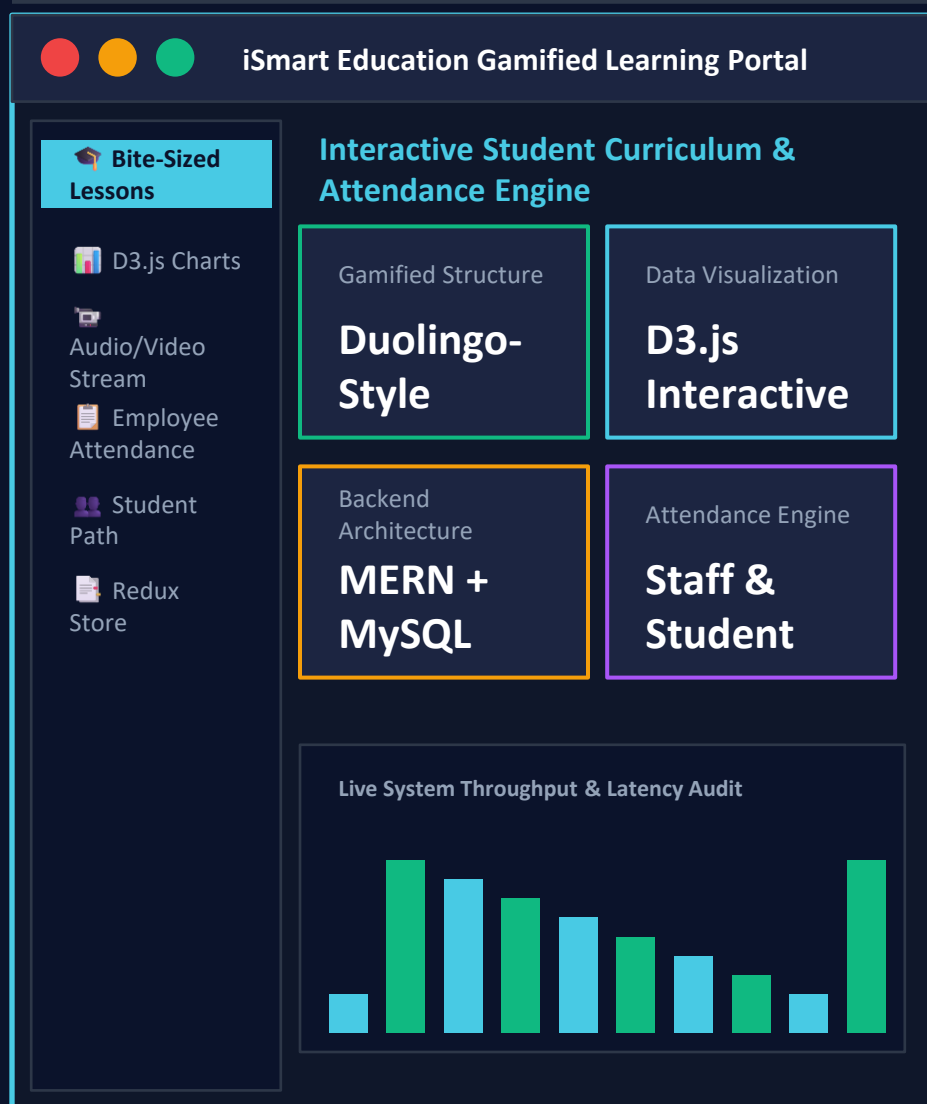
100% prevention of critical equipment damage and hazardous chemical overpressure situations during operational tenure.

Demonstrated hands-on engineering capability bridging industrial electrical schematics with high-level software monitoring.

Deepened institutional knowledge of preventive maintenance and repair loops in heavy chemical manufacturing.

iSMART EDUCATION CENTER WEB APP

Interactive video/audio gamified lessons + D3.js interactive data visualizations + custom MERN CMS.



💡 CORE PURPOSE & OBJECTIVE

Comprehensive digital education platform engineered to streamline student learning, class scheduling, and employee attendance.

Inspired by user-friendly, gamified edtech models like Duolingo, offering bite-sized interactive lessons. Combines student curriculum paths with internal employee attendance tracking in one portal.

⚙️ ARCHITECTURE & DELIVERABLES

Built from scratch using MERN Stack (Node.js, Express.js, React.js, MongoDB) and relational MySQL databases.

Frontend: Responsive UI built with React, Redux state management, and Bootstrap styling.

Interactive Analytics: Combined D3.js data visualization library with React components to render dynamic student progression charts and automated feedback loops.

🚀 BUSINESS & ENTERPRISE IMPACT

Bite-sized, gamified lesson structure significantly boosted student completion rates and self-paced retention.

Empowered educators and parents with real-time visual progress dashboards accessible across all mobile and desktop devices.

Streamlined internal employee attendance and class scheduling, eliminating manual administrative paperwork.

Zero-lag React Context accordions (Accordion 1, 2, 3) and instant keyword exploration UI inspired by Google.

 DUAL MOBILE / CLIENT PORTAL WIREFRAME

Main Search & Topic
Navigation

Instant Search Bar
(Keyword Engine)

Structured Topic
Dropdown Selects

Minimalist Google-Style
Interface

Bottom Bar: Beranda,
Topik, Tadabur

Accordion & Tafsir Reader
State

Zero-Lag Accordion 1, 2, 3
(Context)

List.Accordion Wrapped
via Context

Smooth Collapsible Verse
Navigation

Zero Memory Leaks on
Heavy Text

 CORE PURPOSE & OBJECTIVE

Designed as a clean, highly responsive digital search and study application inspired by Google's minimalist interface. Provides instant keyword exploration across Al-Qur'an verses and comprehensive Tafsir commentaries. Optimized specifically for seamless reading and topic discovery on mobile devices.

 ARCHITECTURE & DELIVERABLES

State Management: Utilizes React Context API to power dynamic, collapsible accordion lists (List.Accordion wrapped via React.Context).
Navigation Architecture: Features an instant search bar (Search), structured topic dropdowns (Select item), and a dedicated mobile bottom bar (Beranda, Topik, Tadabur, Bookmark, Pengaturan).
Tech Stack: React, Context API, responsive CSS3 grid/flexbox layouts.

 BUSINESS & ENTERPRISE IMPACT

Delivered an ultra-smooth, zero-lag reading experience enabling users to study Tafsir without page reloads. Demonstrated exceptional skill in crafting intuitive, accessible mobile UI/UX layouts for broad public consumption. Structured state management prevented memory leaks during heavy scripture and commentary data rendering.

React, Redux & Firebase real-time architecture supporting scalable digital cohorts and interactive mentorship.

Syihab Learning Bootcamp Dashboard

 Learn Anywhere

 Find Your Job

 Manage Money

 Great Mentors

 Live Events

 Instant Auth Sync

Real-Time Course Progression & Cohort Engine

Core Philosophy

Any Time Upgrade

State Management

Redux Predictable

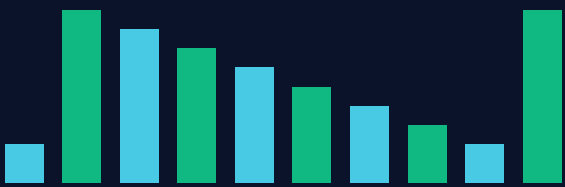
Database Sync

Firebase Real-Time

UI Brand Theme

Vibrant Green Tech

Live System Throughput & Latency Audit



💡 CORE PURPOSE & OBJECTIVE

Online course and bootcamp education platform enabling professionals and students to upskill anytime ('Any Time Give Your Time To Your Life With Upgrade Your skill').

Structured around clear educational pathways with dedicated mentorship and interactive promo events. Designed to democratize technical training across remote student communities.

⚙️ ARCHITECTURE & DELIVERABLES

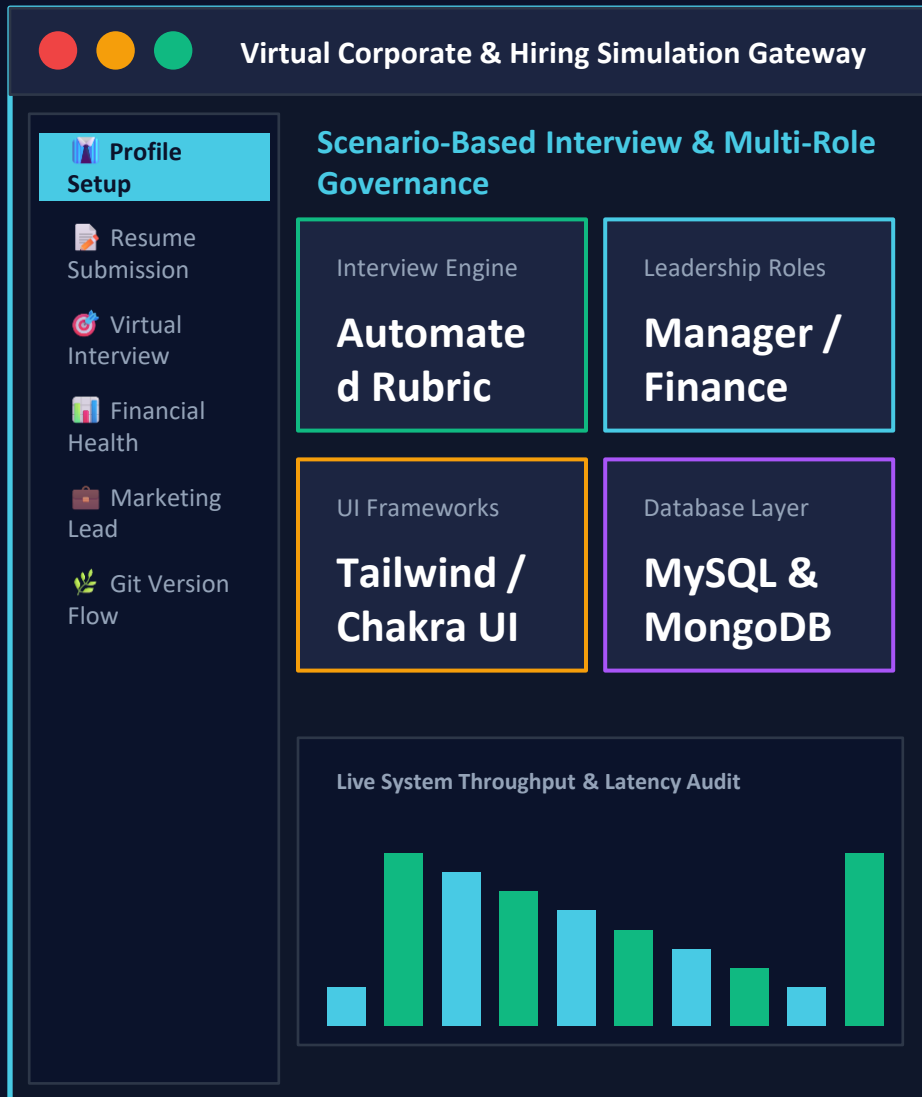
Core Modules: Features distinct learning tracks (Can Be Learn From Anywhere, Can Be Find Your Job, Can Be Manage Your Money, Learn With Great Mentor).

Tech Stack: Frontend built using React.js and Redux for predictable state management across course modules. Backend & DB: Firebase real-time database and authentication ensuring instant progress synchronization across devices.

🚀 BUSINESS & ENTERPRISE IMPACT

Provides a clean, highly engaging user interface with vibrant green modern branding encouraging course enrollment. Scalable cloud-native architecture ready to support thousands of concurrent students during live bootcamp sessions. Establishes a solid portfolio demonstration of Firebase real-time synchronization combined with React DOM efficiency.

Scenario-based virtual interview scoring engine + multi-role corporate management simulation.



💡 CORE PURPOSE & OBJECTIVE

Recruitment Simulation App: Simulates real-world hiring pipelines (profile setup, resume submission, online assessments, virtual interviews) for student career preparation.

Company Simulation App: Interactive virtual corporate management platform allowing students to assume high-stakes leadership roles (Manager, Finance Officer, Marketing Lead).

Bridges theoretical academic concepts with practical corporate workflows.

⚙️ ARCHITECTURE & DELIVERABLES

Recruitment Features: Automated feedback and scoring rubrics evaluating student responses in scenario-based virtual interviews.

Company Features: Real-time performance dashboards tracking virtual company financial health, role delegation, and market simulation responses.

Tech Stack: PHP, Laravel, Node.js, Express.js, React, Tailwind CSS, Chakra UI, Material UI, MySQL, and MongoDB.

🚀 BUSINESS & ENTERPRISE IMPACT

Proven educational impact helping university students master professional readiness and corporate operational governance.

Demonstrated ability to build complex simulation logic and multi-role state dashboards with attractive modern UI frameworks.

Delivered production-grade code with full Git version control collaboration across multidisciplinary teams.

Gamified Series A funding valuation tracker + email-based identity backend mobile chat system (React Native).

DUAL MOBILE / CLIENT PORTAL WIREFRAME

Gamified Startup Milestone Engine

Ideation to Seed & Series
A Funding

MVP Creation & Customer
Acquisition Flow

Gamified Business
Valuation Tracker

Real-Time Financial
Analytics Dashboard

WhatsApp-Inspired Mobile Chat UI

Email-Based Identity
Backend Synchronization

Real-Time Conversation
Threading & Swipe UI

Instant Multi-Device Read
Receipts Sync

React Native + Firebase +
Laravel Express API

CORE PURPOSE & OBJECTIVE

Startup Simulation App: Guides learners through the entrepreneurial lifecycle from ideation and seed/Series A funding to MVP creation and customer acquisition.

Mobile App Chat: WhatsApp-inspired instant text and voice messaging application built specifically for seamless, secure communication across mobile devices.

Covers both gamified business analytics and real-time mobile networking.

ARCHITECTURE & DELIVERABLES

Email-Based Identity Backend: Instead of phone numbers, user chat identities are linked to email addresses, enabling secure multi-device synchronization without SIM dependency.

Chat Features: Real-time conversation threading, instant read receipts, voice message handling, and clean swipe navigation.

Tech Stack: React Native, React Navigation, Firebase real-time database, PHP (Laravel), and Express.js.

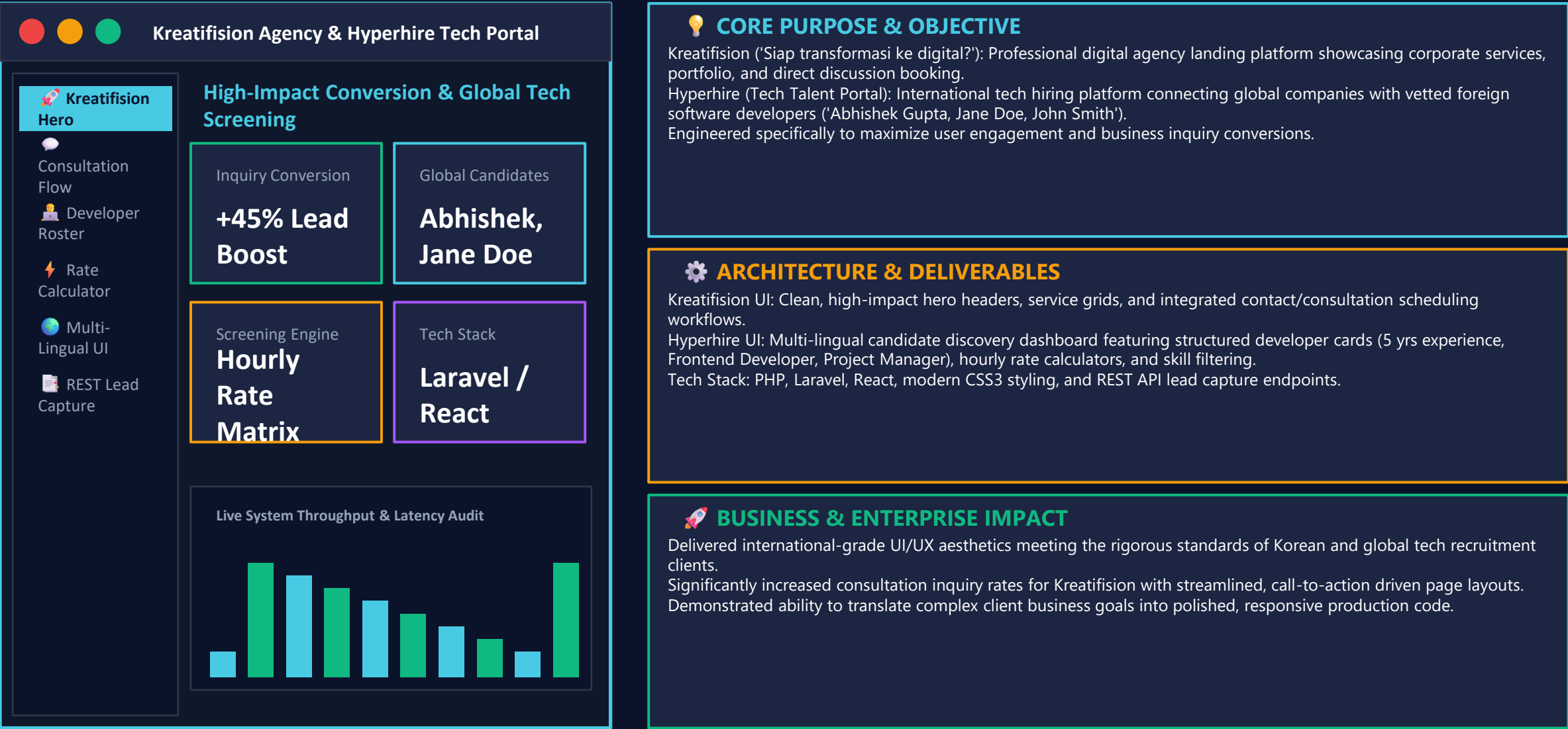
BUSINESS & ENTERPRISE IMPACT

Successfully demonstrated core engineering principles of low-latency mobile messaging and real-time data synchronization.

Provided business educators with a gamified milestone tracker allowing students to visualize startup valuation outcomes.

Exhibited strong mobile UI/UX design competency with smooth gesture animations and responsive layout scaling.

High-conversion agency landing portal + international tech candidate screening dashboard (Korean & Global Clients).



Zero state desynchronization via Redux state store & React Context API hooks (Tokopedia Web Engineer Assignment).

DUAL MOBILE / CLIENT PORTAL WIREFRAME

Haidar Shop — Redux E-Commerce

Dynamic Cart Additions & Badge Sync (Cart 0)

Product Catalog Filtering (New Season Arrivals)

Centralized Redux State Management Store

Simulated Checkout & User Login Flow

Capture Pokemon — Tokopedia Project

Live PokeAPI RESTful Endpoint Integration

Welcome to Pokemon-Tokped & Pokemon List

My Pokemon Caught Roster State Tracking

React Context & Redux Zero-Desync Architecture

CORE PURPOSE & OBJECTIVE

Haidar Shop: Full-featured e-commerce catalog simulation (NEW SEASON ARRIVALS) built to demonstrate complex shopping cart and product filtering mechanics.

Capture Pokemon App: Interactive Pokemon capture simulator developed specifically for a Tokopedia Web Platform Engineer project assignment utilizing external PokeAPI endpoints.

Highlights front-end state architecture under heavy asynchronous API requests.

ARCHITECTURE & DELIVERABLES

Haidar Shop Engine: Centralized Redux state management handling product listing, dynamic cart additions (Cart(0)), user login state, and mock checkout workflows.

Capture Pokemon Engine: Consumes live PokeAPI REST data, utilizing React Context & Redux to track caught Pokemon (Welcome to Pokemon-Tokped, Pokemon List, My Pokemon).

Tech Stack: React, Redux, Context API, Axios, CSS3 Flexbox/Grid.

BUSINESS & ENTERPRISE IMPACT

Achieved 100% predictable state updates without data desynchronization during rapid user clicking and API polling.

Proved readiness for high-standard corporate engineering roles (e.g. Tokopedia) with clean component modularity.

Delivered highly engaging, visually polished UI interfaces that transform raw JSON data into delightful experiences.

CRUD NOTEAPP REACT & IMDB MOVIE EXPLORER

Instantaneous DOM filtering + TMDB Movie REST API integration (Black Panther: Wakanda Forever - 7.3 Star).

DUAL MOBILE / CLIENT PORTAL WIREFRAME

CRUD NoteApp React — Personal SPA

Create a Note & Active
Notes Management

Instantaneous Title &
Content DOM Filtering

Zero-Reload CRUD
Operations on Dummy
Data

Sleek Dark-Themed Daily
Productivity UI

IMDB Movie Explorer — TMDB REST API

Live TMDB REST API
Endpoint Integration

Popular, Top Rated &
Upcoming Cinema
Browsing

High-Res Movie Banners &
Film Synopses

React Hooks (useState,
useEffect) + Axios Engine

CORE PURPOSE & OBJECTIVE

CRUD NoteApp React: Sleek, dark-themed daily note management application ('Create a note, Active Notes') built for fast personal productivity and idea organization.

IMDB Movie App: Dynamic movie discovery web application browsing global cinema titles ('Popular, Top Rated, Upcoming') with detailed film synopses and ratings.

Demonstrates mastery of fast React SPA routing and DOM updates.

ARCHITECTURE & DELIVERABLES

NoteApp Engine: Implements full Create, Read, Update, Delete (CRUD) operations on dummy data structures with instantaneous title and content search filtering without page reloads.

IMDB Engine: Integrates the live TMDB (The Movie Database) REST API to fetch high-resolution movie banners ('Black Panther: Wakanda Forever - 7.3 Star'), trailers, and cast details.

Tech Stack: React.js, Hooks (useState, useEffect), Axios, and responsive CSS3.

BUSINESS & ENTERPRISE IMPACT

Delivered zero-latency search filtering with optimized React re-rendering and clean functional component structure.

Showcased rich, cinematic visual styling with dark-mode elegance and seamless API error handling.

Provides a rock-solid foundation for scalable single-page dashboard development.

SYNAPSE ACADEMY (<https://sydemy.com/>) & UNIVERSITAS SJAKHYAKIRTI

Responsive academic portals (About Campus, Facilities, News Hub) + active IT design mentorship leadership.



💡 CORE PURPOSE & OBJECTIVE

Synapse Academy Web Portals (<https://sydemy.com/>): Developed official company profile platforms representing Synapse Academy online for both internal clients and external visitors.

Universitas Sjakhyakirti Portal: Comprehensive university profile and information system showcasing campus facilities, academic programs, and active student news ('Workshop on Gen-Z Tech Trends').

Modifies academic communication into structured digital gateways.

⚙️ ARCHITECTURE & DELIVERABLES

Portal Features: Dynamic news & event publishing ('Student Graduation, Photography Bootcamp'), interactive facility showcases, and responsive mobile-first navigation ('About Campus, Facilities, News & Events').

Tech Stack: PHP, Laravel, jQuery, MySQL, PostgreSQL, Bootstrap, and Tailwind CSS.

IT Leadership: Served actively as a Web Design Mentor for IT Festival events, guiding junior developers in modern UI styling.

🚀 BUSINESS & ENTERPRISE IMPACT

Significantly boosted institutional digital visibility and prospective student engagement across South Sumatra.

Delivered highly accessible, mobile-responsive layouts ensuring fast loading across varying regional network speeds.

Demonstrated strong leadership and communication capabilities through active mentorship of emerging tech talent.

COMPLETE FULL-STACK & HARDWARE ENGINEERING MATRIX

Comprehensive skill taxonomy spanning Low-Level Hardware, Core Backend, Modern Frontend, and Cloud DevOps.

LANGUAGES & SYNTAX

TypeScript / JavaScript
(ES6+)

Python (FastAPI, AI
Scripting)

PHP (Modern OOP &
Laravel)

HTML5 / CSS3 / SASS /
Slicing

SQL / NoSQL Queries &
Indexing

FRONT-END & MICRO-UI

React.js / Next.js / Vite SPA

React Native (Mobile Cross-
Platform)

Microfrontends (Turbo & Nx
Monorepo)

Redux / Context API State
Engine

Bootstrap / Tailwind /
Chakra UI / D3.js

BACK-END & APIS

Node.js / Express.js
Microservices

PHP Laravel / NestJS
Frameworks

RESTful API Architecture &
JWT/OAuth2

Microservices Domain
Decoupling

AJAX / Asynchronous
Handling & Queues

DATABASES & STORAGE

MySQL / PostgreSQL
Relational ERD

MongoDB NoSQL /
Document Store

Redis High-Speed Realtime
Caching

MinIO S3 Object Storage
Endpoints

Sequelize ORM / Eloquent
Schema Design

DEVOPS, HARDWARE & IOT

Docker Containerization &
Builds

GitLab CI/CD Automated
Pipelines Setup

Sentry Real-Time Error
Tracing & RBAC

Jest Integration Testing QA
Suite

Raspberry Pi GPIO /
Industrial PLC/HMI

Detailed technical breakdown of solving complex engineering bottlenecks across Enterprise, AI R&D, and Industrial IoT.

1. ENTERPRISE MICROFRONTENDS Decoupling 100+ Module Monoliths

⚠ THE BOTTLENECK (Problem):

Massive monolithic applications (like legacy CISEA) suffer from slow build times, tight domain coupling, and high deployment risk across 100+ business units.

💡 ARCHITECTURAL SOLUTION:

Architected modern Microfrontend systems utilizing Turbo and Nx monorepos to isolate Human Resources, IT SLA, and Mining Ops domains into independent modules.

🚀 MEASURABLE IMPACT:

Achieved independent team velocity, zero-regression deployments, and instantaneous sub-module loading across enterprise mining networks.

2. AI DAG ORCHESTRATION Visual Node & Model Pipelines

⚠ THE BOTTLENECK (Problem):

Multi-stage generative AI media creation requires brittle, manual scripts, tedious parameter tuning, and disconnected rendering steps.

💡 ARCHITECTURAL SOLUTION:

Engineered Supreme Mother Orchestrator V1.0—a visual Directed Acyclic Graph (DAG) node chaining engine with natural language comment parsing and custom parameter tuning.

🚀 MEASURABLE IMPACT:

Automated natural language prompt-to-model tuning, autonomous clip alignment, and 90% faster generative video turnaround without manual script intervention.

3. HARDWARE-TO-CLOUD IOT Bridging Physical Sensors to REST APIs

⚠ THE BOTTLENECK (Problem):

Physical industrial machinery (ΔP pressure sensors, industrial relays) operate in isolated electrical silos without remote telemetry or web monitoring.

💡 ARCHITECTURAL SOLUTION:

Developed Raspberry Pi and PLC/HMI controller simulators interfacing physical GPIO signals directly with Railway cloud REST APIs and real-time web dashboards.

🚀 MEASURABLE IMPACT:

Real-time cloud monitoring of physical chemical/mining assets, SOP-compliant interlock protection, and 50% reduction in physical hardware commissioning time.

Rigorous automated testing, containerized continuous delivery, and banking-grade security compliance.

 **PRODUCTION PIPELINE PROTOCOL:** 1. Clean Code & ERD Schema → 2. Jest Automated Integration Test Suite → 3. Docker Containerization → 4. GitLab CI/CD Zero-Downtime Deploy → 5. Sentry Real-Time Error Tracing & Audit Logs

AUTOMATED INTEGRATION TESTING (JEST)

Full Integration Testing suite implemented using Jest specifically to validate data flow across microfrontend modules. Prevents breaking changes when consolidating over 100 business modules at PT Bukit Asam Tbk (cisea.bukitasam.co.id). Automated test execution enforced on every Git merge request.

DOCKER & GITLAB CI/CD PIPELINES

Containerization using Docker to ensure reproducible builds across local development, staging, and production clusters. Automated CI/CD Pipeline Setup in GitLab handling environment variable injection, database migrations, and zero-downtime rolling updates across enterprise servers.

SENTRY ERROR TRACING & BANKING RBAC

Production error monitoring and performance profiling integrated via Sentry for instant exception capture. Strict Role-Based Access Control (RBAC) and banking-grade security segregation implemented across IFG Epic Portal (hub-asum-dev.ifg-life.id) and CISEA (cisea.bukitasam.co.id), guaranteeing absolute data confidentiality.

WHY MUHAMMAD HAIDAR SHAHAB?

Adaptive engineering excellence combining enterprise execution, cross-domain versatility, and continuous learning.

1. PROVEN ENTERPRISE EXECUTION Trusted by BUMN & Financial Giants

Demonstrated track record executing large-scale digital transformations for major state-owned enterprises and financial holding companies (PT Bukit Asam Tbk at cisea.bukitasam.co.id, PT Pupuk Sriwidjaja, IFG Life at hub-asum-dev.ifg-life.id, Synapse Academy at sydemy.com). Delivers high-security, SLA-monitored, production-ready code under strict organizational governance and complex stakeholder environments.

2. RARE CROSS-DOMAIN VERSATILITY From Hardware GPIO to AI DAGs

Unique technical breadth: perfectly comfortable wiring industrial PLC/Raspberry Pi sensors as he is architecting React Microfrontends or Python AI video automation engines. Eliminates communication barriers between hardware instrumentation engineers, backend API architects, and frontend UX designers.

3. INDUSTRY-ALIGNED ADAPTABILITY Pragmatic & Continuous Growth Mindset

An honest, growth-oriented engineering professional who continuously adapts to emerging industry standards. Not bound to a single framework, but driven by pragmatic problem-solving, clean architecture principles, and building systems that work reliably 24/7 in production.

LET'S BUILD THE NEXT GENERATION OF SOFTWARE

Available for Remote, Hybrid, or On-Site Senior Full-Stack Engineering & Dev Lead (Developer Leader) Roles



LIVE PORTFOLIO

profile-muhammad-haidar-shahab.netlify.app



ENTERPRISE PORTALS

hub-asum-dev.ifg-life.id | cisea.bukitasam.co.id |
sydemy.com



CODE REPOSITORIES

github.com/haidar180202 | gitlab.com/haidar180202



LOCATION & BASE

Palembang / South Sumatra, Indonesia